

AML Transaction Monitoring Analysis

Project Overview:

This project analyses 2000 clients and 50,000 transactions for FATF/OFAC exposure and transaction-monitoring alerts. Here the client master is joined with the transaction ledger to quantify alert activity, identify concentration by sector and country, measure monthly stability, and rank clients for review. A transaction is labelled **suspicious** when at least one of five transaction-level flags is active - OFAC match, FATF-country exposure, structuring, rapid movement, or trade mispricing.

Key Performance Indicators:

- **50,000 transactions** across **2,000 clients**
- Total transaction value of approximately **\$209.8 million**
- **4,489 suspicious transactions**, representing **8.98%** of all transactions
- Suspicious transactions represented approximately **\$22.2 million**, or **10.6%** of total transaction value.
- Average transaction value is determined to be \$ 4,194.14.
- Median Transaction Value is \$1582.19.
- Average transaction value is approximately **2.65 times the median transaction value**, i.e., Mean-to-Median Ratio, which supports the conclusion that transaction values are **right-skewed**.
- High Risk clients are estimated to be 1,218 and High-Risk client Transactions are estimated to be 36,370.

Key Findings

- **Rapid movement is the dominant alert-** It appears in 2,460 transactions (4.92%), followed by OFAC transaction matches in 1,790 transactions (3.58%). These two rules account for the majority of flagged events which is approximately **81.6%** of transaction-level rule triggers and should be the first focus for alert-quality review.
- **Energy/Oil** with 10.67% had the highest suspicious transaction rate among sectors. Defence/Arms and Import/Export are also among the higher-rate segments, while Real Estate and Casino/Gambling show lower rates in this sample. Tech is labelled Low risk but runs at **9.20%**. Rates should be interpreted with volume alongside them.
- **Venezuela**, represented by country code **VE**, has the highest suspicious rate at 12.71% across 2,722 transactions among all client countries and with all 108 clients OFAC-flagged. **North Korea (KP)** and **Iran (IR)** show 100% FATF and OFAC flag rates in the observed population. This is a prioritization signal, not proof that all activity from that country is illicit.
- **The transaction amount distribution** is strongly right-skewed. The mean amount is \$4,196.14, while the median is \$1,582.18, indicating that a relatively small number of large transactions materially influence total value. This supports using both count-based and value-based monitoring thresholds.

- **Suspicious rates by transaction type** are: CHECK 9.34%, WIRE 9.13%, SWIFT 8.84%, ACH 8.60%. The differences are useful for segment-specific rule tuning, but they should be tested for statistical significance and operational cost before changing thresholds.
- **1,218 High-risk clients** are 60.9% of the base; their suspicious rate is 11.60%, versus 4.99% for Medium and 1.53% for Low. PEP clients average **\$6,365** per transaction versus **\$4,063** for non-PEPs; suspicious rates are 10.81% versus 8.87%.
- Repeated 100%-**suspicious corridors** involve Venezuela, Syria, Russia, and Iran. Domestic activity is smaller but slightly more alert-dense than cross-border activity: 10.00% versus 8.93%.
- The **\$5,000–\$9,999** band has the highest suspicious rate, **15.43%**; near-\$10,000 clustering warrants review.
- A transparent client-prioritization score was created using suspicious transaction counts, alert counts, and transaction value. Jones-Atkinson has highest **\$154.9 K** outlier transaction versus a **\$ 7.1 K** client average, z-score **6.18**.

Recommendations:

Operational controls:

Review rapid-movement alerts for false-positive patterns, especially where transfers are routine treasury activity. Introduce a second-stage rule that considers velocity, amount, counterparty novelty, and time between outgoing and incoming movements.

Risk-based segmentation:

Apply enhanced due diligence and tighter monitoring to clients in Energy/Oil, Defence/Arms, and Import/Export, while preserving volume-based controls so that lower-rate but high-value segments are not overlooked.

Alert quality and model governance:

Capture investigator outcomes for every alert, then measure precision, false-positive rate, time to-disposition, and value-at-risk. These labels can support a calibrated scoring model and threshold optimization.